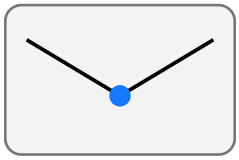
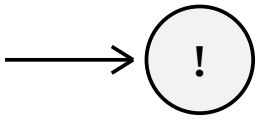
SOCIAL ENGINEERING ATTACKS & MAJOR SECURITY BREACHES

GMIU Cybersecurity Infographic Assignment





PHISHING
Fake message → suspicious link → sensitive information at risk

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Subject: Cybersecurity / CNS
Purpose: Explain real-world social engineering risks, attack techniques, case studies, and prevention.

1. Introduction and Real-World Problem

Social engineering is a cyberattack technique in which criminals manipulate people into sharing confidential information, clicking suspicious links, downloading unsafe files, or granting unauthorized access. It targets human trust, fear, curiosity, and urgency rather than relying only on technical weaknesses.

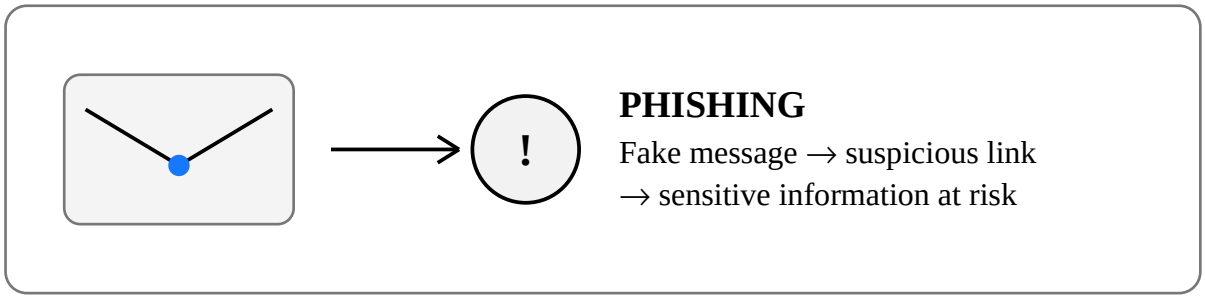
Problem statement: People use online banking, UPI, social media, email, and educational websites every day. Attackers exploit this dependence through fake messages, impersonation, and tempting offers. These scams can lead to financial loss, identity theft, stolen accounts, and data breaches.

Common targets and objectives

Targets include students, bank customers, employees, IT support staff, and organizations. Attackers may aim to steal passwords, obtain financial information, install malware, access private systems, or expose confidential data.

2. Main Social Engineering Techniques

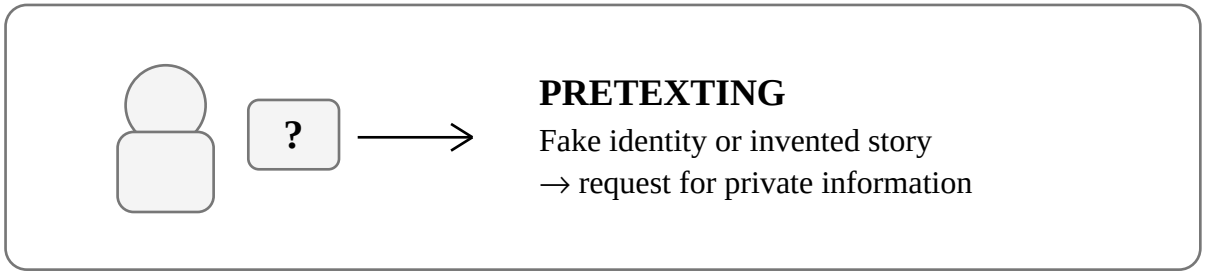
A. Phishing



Phishing uses fake emails, SMS messages, or websites that appear to come from a trusted source. The victim may be asked to verify an account, claim a prize, or open an attachment. The goal is often to steal login details or encourage an unsafe action.

Example: A message claims that a bank account will be blocked unless the user verifies information through a link. **Prevention:** Check the sender and website address; avoid unexpected links and attachments; visit the official website or app directly.

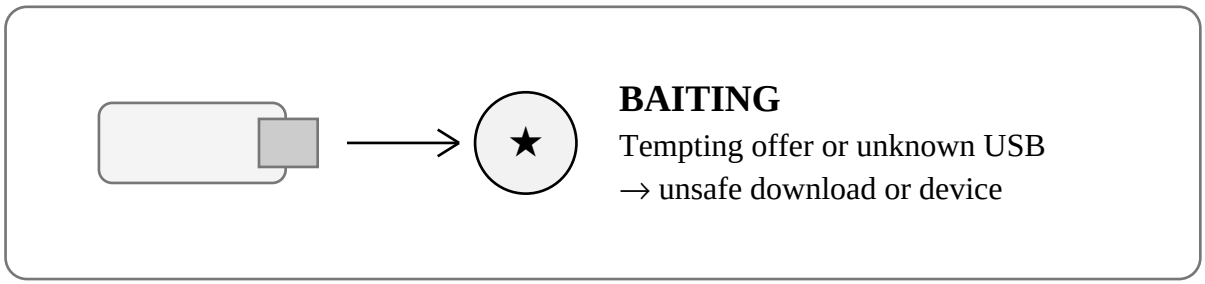
B. Pretexting



Pretexting is when an attacker invents a believable story or pretends to be a trusted person, such as a bank representative or college IT employee, to obtain confidential information. Verify identity using an independent official channel, and never share passwords or OTPs.

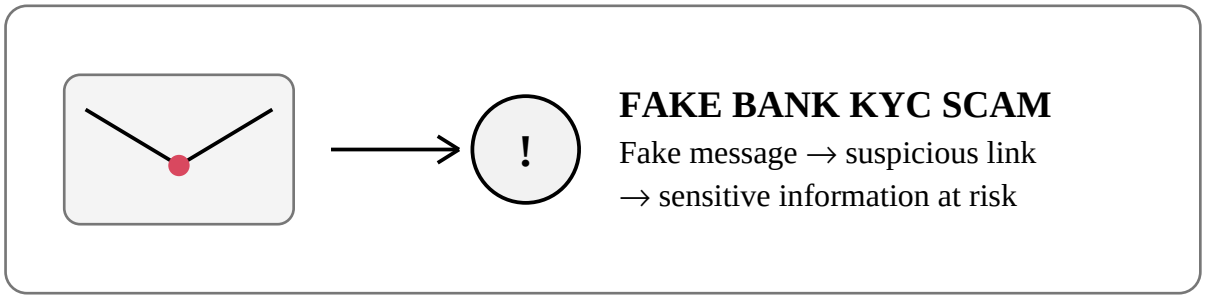
2. Main Techniques (continued)

C. Baiting



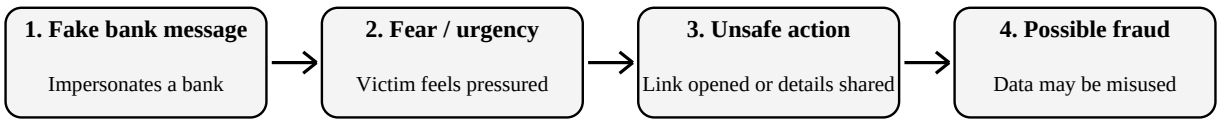
Baiting attracts victims with something tempting, such as free software, a prize, or an unknown USB drive. The victim may download a harmful file or connect an unsafe device. Download software only from trusted sources and never connect unknown USB devices.

3. Real-World Problem: Fake Bank KYC Scam



Fraudsters may impersonate a bank and send an urgent message claiming that KYC must be updated immediately or the account will be blocked. A link may lead to a fraudulent website that attempts to collect banking details. If those details are misused, the victim may suffer financial loss or account compromise.

Attack flowchart



Root cause: Trust in a message that appears official. **Weakness exploited:** Fear, urgency, and lack of awareness. **Possible impact:** Financial fraud, identity theft, and privacy loss.

Prevention: Contact the bank using its official app, website, or phone number. Never share OTPs, UPI PINs, or passwords. Do not use contact details supplied in a suspicious message.

4. Major Case Study: MGM Resorts (2023)

MGM Resorts International experienced a major cybersecurity incident in September 2023. Public reporting indicated that social engineering was used to help attackers gain initial access to systems. The incident disrupted some hotel and casino services, and MGM reported that personal information belonging to some customers was obtained.

Impact

• Disruption to hotel and casino operations.
• Problems with some digital services and reservations.
• Exposure of some customer personal information.
• Financial costs and potential reputational damage.

Analysis: What could reduce the risk?

• Verify identity before responding to IT support or account-recovery requests.
• Use multifactor authentication and tightly controlled access.
• Train employees to identify impersonation and suspicious requests.
• Monitor unusual activity and prepare an incident-response plan.
• Maintain secure backups and test recovery procedures.

Lesson learned

Even large organizations can be vulnerable when attackers exploit human trust. Public accounts identify social engineering as part of the incident, but not every technical step is fully established in public reporting.

5. Another Example: Twitter Account Compromise (2020)

Attackers used social engineering against employees to gain access to internal tools and take over prominent accounts. The incident highlighted the need for employee awareness, restricted administrative access, monitoring, and strong internal security controls.

6. Prevention Strategies

Verify identity Confirm requests for money or confidential information through official channels.	**Protect credentials** Use strong, unique passwords and enable multifactor authentication.
Check links Inspect website addresses and avoid unexpected attachments or downloads.	**Build awareness** Learn to recognize phishing, impersonation, and manipulative messages.
Limit access Organizations should give each user only the permissions they need.	**Report quickly** Contact the bank, college IT team, or relevant cybercrime reporting service.